



# Final Report

*An AI-Powered Medication Adherence & Drug-Interaction Copilot*

Management of AI Products (MBACC ZG583)

2024BC26670 Sangitha V

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# 1. Product Strategy and Opportunity Framing

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## Product Vision

Become the trusted layer between what a doctor prescribed and what a patient actually takes, reducing preventable harm from managing multiple chronic medications. The product starts as a consumer companion app and is designed from day one to be embedded as a feature inside an existing pharmacy, insurer, or hospital app rather than compete as a standalone destination.

## Jobs-to-Be-Done

- When I get a new prescription, help me understand quickly and calmly whether it is safe with what I already take, so I do not have to call three different doctors to find out.
- When I am juggling multiple daily medications, remind me in a way that fits my routine, so I do not forget a dose or double up.
- When something looks risky, tell me clearly what to do next and who to talk to, so I feel supported, not just alarmed.

## Opportunity Framing

Why this problem matters: missed doses and unrecognized drug interactions are a leading, preventable cause of hospital readmission among patients on three or more chronic medications, and no single caregiver in the current system has continuous visibility into what a patient is actually taking.

Why current alternatives are weak: a physician checks interactions once, against what they know about, at the moment of prescribing. A pharmacist checks again at pickup. Neither sees what a patient adds afterward such as an over-the-counter painkiller, a supplement, a second prescription from a different specialist on a different record. A plain reminder app solves adherence but says nothing about interaction risk; a static drug-lookup tool has no adherence context and no plain-language explanation a patient will actually trust.

Why AI may help: the interaction-checking logic itself is a reliable, rules-based lookup, not a hard AI problem. What genuinely benefits from AI is turning a structured, sometimes frightening piece of clinical information into language a patient can act on calmly, without needing a medical degree to interpret it.

## Opportunity Matrix

| Idea                                       | Business Value   | Feasibility            | Priority |
|--------------------------------------------|------------------|------------------------|----------|
| Rules-based interaction + adherence engine | High             | High                   | P0       |
| GenAI plain-language explanation layer     | Medium           | Medium                 | P1       |
| Personalized adherence-risk prediction     | High             | Low (needs pilot data) | P2       |
| EHR / pharmacy system integration          | High (long-term) | Low                    | P2       |

## Build vs. Buy vs. Partner/API

Recommended path: buy/API the underlying drug data (DrugBank, RxNorm, openFDA); build the interaction-reasoning and risk-scoring layer in-house; buy the language model (a general-purpose LLM API) but build the clinical-safety prompt and human-review workflow around it in-house; partner for distribution through health plans, PBMs, and pharmacy chains.

Why: the differentiated part of this product is not the underlying drug data, which is a freely available commodity, or the raw language model, which is also a commodity. It is the safety-scoring logic and the human-review workflow wrapped around it. Building a pharmacological database from scratch, or training a bespoke model, would spend engineering effort on an already-solved problem instead of the actual differentiator.

Main trade-offs: relying on licensed or free data sources rather than proprietary data means the moat has to come from workflow and trust design, not data exclusivity. Depending on a general-purpose LLM API exposes the product to that vendor's pricing and model-behavior changes over time, which are mitigated by keeping the actual safety decision outside the model entirely, in the deterministic rules engine.

## First Wedge

Start narrow: a single condition (Type 2 diabetes with common cardiometabolic comorbidities such as hypertension, high cholesterol), one language, and a focused reference set of frequently co-prescribed drugs. Prove the pharmacist-agreement gate on that narrow slice before expanding to additional conditions or languages.

## 2. PM Artifact Pack

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### Problem Statement

Patients managing three or more chronic prescriptions frequently miss doses or unknowingly combine drugs in ways that cause harm, not because they are careless, but because no one in the current system has continuous visibility into their full, current medication list.

### PRD Summary

- User problem: stated in the *Jobs-to-Be-Done* section.
- Core workflow: capture a medication → check it against the current list → add it safely or flag a risk → a pharmacist reviews anything moderate or high risk before the patient sees it → the patient checks in daily on adherence.
- AI role: a deterministic rules engine handles the safety-critical classification; a generative layer (introduced as a Phase 2 preview) turns a flag into plain language, always kept behind a pharmacist for anything above low risk.
- Risks: see Section 6, Ethics, Governance, and Risk Note.
- Fallback logic: any medication the reference dataset does not recognize is never treated as safe by default, it is explicitly routed to a human pharmacist for review, the same fallback used for anything the interaction engine cannot confidently classify.

## Feature Prioritization

| Feature                                             | User Value | Effort | Confidence       | Priority         |
|-----------------------------------------------------|------------|--------|------------------|------------------|
| Barcode/photo medication capture                    | High       | Medium | High             | P0 — MVP         |
| Interaction alert (rules-based + pharmacist review) | High       | Medium | High             | P0 — MVP         |
| Daily adherence check-in                            | Medium     | Low    | High             | P0 — MVP         |
| GenAI plain-language explanation                    | Medium     | Medium | Medium           | P1 — Fast follow |
| Caregiver dashboard                                 | Medium     | Medium | Medium           | P1               |
| Personalized adherence-risk prediction              | High       | High   | Low (needs data) | P2 — Scale       |
| EHR / pharmacy system integration                   | High       | High   | Low              | P2 — Scale       |

## MVP Definition

The MVP is successful if it can reliably flag known interactions for a narrow drug set with zero false negatives, hold every moderate- or high-risk case for pharmacist review rather than auto-approving it, and log daily adherence without needing any live AI call to function.

## Roadmap

| Phase                        | Timeframe   | Focus                                                                                | Key Milestone                                   |
|------------------------------|-------------|--------------------------------------------------------------------------------------|-------------------------------------------------|
| Phase 1: MVP                 | Months 1–4  | Rules-based interaction engine + adherence reminders, single condition, one language | 60-day pilot; $\geq 95\%$ pharmacist agreement  |
| Phase 2: GenAI & expansion   | Months 5–8  | Plain-language explanations for low-risk alerts; multi-condition; multilingual UX    | False-positive rate $< 5\%$ on auto-sent alerts |
| Phase 3: B2B distribution    | Months 9–14 | PBM/health-plan pilot; caregiver dashboard; formal regulatory review                 | $\geq 1$ signed PBM/insurer pilot               |
| Phase 4: Scale & integration | Months 15+  | EHR/pharmacy API integration; personalized risk prediction                           | Embedded in $\geq 1$ partner platform           |

## Success Metrics

- Product metric: medication adherence rate  $\geq 85\%$  at 90 days.
- Business metric: B2B pilot-to-contract conversion  $\geq 30\%$  of pilots.
- Trust / guardrail metric: false-positive interaction rate  $< 5\%$ ; pharmacist agreement  $\geq 95\%$  sustained over a 60-day pilot.

## 3. UX, Workflow, and Trust Design

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### Main Workflow

A patient scans or searches a medication into the app. The system checks it against their current list using a real, FDA-grounded interaction reference. If there is no known risk, the medication is added immediately. If there is a moderate or high risk, the patient sees a plain-language explanation and the item is simultaneously routed to a pharmacist review queue. If the medication is not recognized at all, the patient is told it is not yet in the reference set, and it is routed to the pharmacist rather than silently treated as safe. Once a day, the patient taps Taken, Skipped, or Remind me later, a single low-friction interaction, not a form.

### Wireframes / Screens

Wireframes for a screen-by-screen breakdown of what was actually built in MVP 1: the medication list and add-medication flow, the interaction alert card, the pharmacist review queue, and the KPI dashboard are added in the *Appendix*.

### Trust and Explainability

- What the user needs to know: that an alert comes from real, cited data (for example, “Based on FDA label data”), not an unattributed AI claim, and that the app does not replace a doctor or pharmacist.
- What must be visible: the risk band (Low / Moderate / Talk to your pharmacist), the source of an interaction claim, and a persistent one-tap path to a human.
- What should be hidden: raw probability scores or model confidence numbers, which help patients to act on clear guidance, not a percentage they have no clinical context to interpret.

### Uncertainty Handling

- High-confidence behavior: shown directly, with a clear risk band and plain-language explanation.
- Low-confidence behavior: routed to the pharmacist review queue rather than shown directly to the patient.
- Failure or fallback behavior: an unrecognized medication is never treated as safe by default, it is always routed to a pharmacist.

### Human-in-the-Loop

Every moderate- or high-risk interaction alert is held in a pharmacist review queue before it reaches the patient; the pharmacist can approve it to the patient or dismiss it as a false positive. This is not a UI convention layered on top of an automated decision, it is implemented and covered by an automated test as the actual gate inside MVP 1’s interaction engine.

## 4. Data, Model, and Evaluation Strategy

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### Problem as an AI Task

Two distinct tasks, not one model: a classification/lookup task (does this combination of drugs carry a known interaction risk?) handled by a deterministic rules engine, and a generation task (turning a risk flag into a

plain-language explanation) introduced as an opt-in Phase 2 preview, kept separate from the safety-critical path.

Data Sources

- Internal: first-party adherence and daily check-in history, collected with explicit consent.
- External: DrugBank, RxNorm (NLM), and openFDA (all free, structured, and API-accessible today).
- User-generated: photo or barcode scans of medication labels.

Labeling / Ground Truth

Interaction severity in the current reference table is a small, manually curated illustrative starting set. A production version needs labels sourced from a licensed clinical database such as DrugBank, rather than hand-authored pairs, fabricating interaction data at scale is not something that can be done honestly for a patient-safety feature, so coverage was deliberately extended using real openFDA label text instead of invented pairwise data.

Baseline

The baseline this product is measured against is not “no product”, it is the status quo of point-in-time prescriber and pharmacist checks with no continuous visibility. The bar to clear is catching what that baseline structurally misses, not outperforming a sophisticated existing system.

Model / API Choice

A deterministic, rules-based lookup handles the safety-critical classification enabling no model, no training-data risk, and fully reviewable by a human. A general-purpose LLM API (Claude) handles the plain-language explanation layer, chosen over training a bespoke model because language generation for this use case does not need domain-specific fine-tuning, it needs a well-designed prompt and a human safety net behind it.

Evaluation Framework

| Metric Type       | Metric                                                 | Why It Matters                                                                            |
|-------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Technical         | Pharmacist agreement with flagged risk level           | Confirms the rules engine’s classifications match real clinical judgment                  |
| Business          | Medication adherence rate; interaction alerts acted on | Confirms the product changes behavior, not just generates alerts                          |
| Trust / Guardrail | False-positive rate; pharmacist review turnaround      | Confirms alert fatigue is not eroding trust and the human-review step is not a bottleneck |

Launch Thresholds

≥ 95% pharmacist agreement with the flagged risk level, sustained over a 60-day pilot, and a false-positive rate under 5%, before any expansion beyond the initial narrow condition and drug set.

# MedGuardian AI — MVP 1 System Architecture

Config-driven, layered, deterministic — zero AI/LLM calls in this build (see left panel for where AI plugs in at Phase 2)

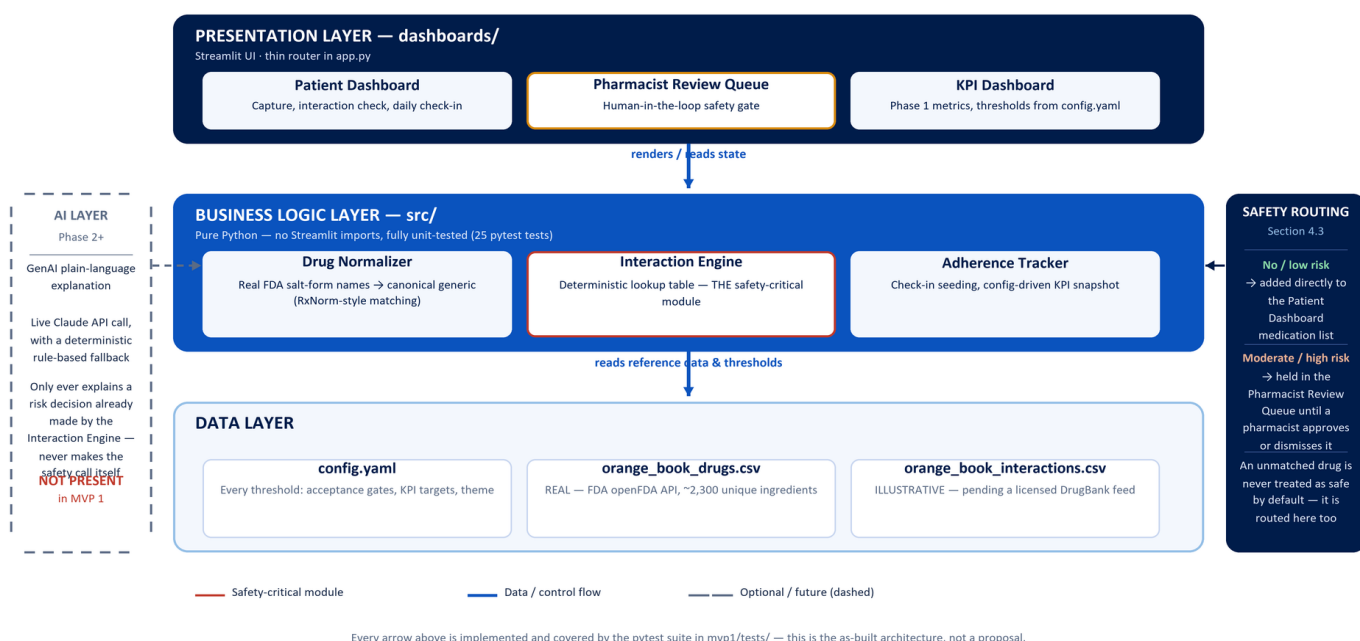


Figure 1. MVP 1 system architecture — presentation, business-logic, and data layers, with the safety-routing branch and the Phase 2 AI layer shown as a separate, opt-in panel.

## 5. Business, Economics, and Scaling View

### Monetization Model

| Model                            | Description                                                                                                                                                                 |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B2B2C — PMPM licensing (primary) | Health plans and PBMs pay a per-member-per-month fee to offer the product as part of a chronic-care management program, justified by measurable readmission-cost avoidance. |
| Direct-to-consumer freemium      | Core adherence reminders and low-risk interaction checks are free; a premium tier (caregiver dashboard, unlimited pharmacist chat) is a paid subscription.                  |
| API licensing                    | Pharmacy chains and EHR vendors license the interaction/adherence engine as an embedded feature, priced on usage volume.                                                    |
| Specialty Pharma PSP program     | A flat annual platform fee or a per-enrolled-patient fee, sold to a pharma manufacturer's Patient Support Program team for a specific specialty or biologic drug.           |

## Value Logic

The product creates value by helping prevent avoidable hospital readmissions and specialty-drug discontinuations; both of which cost the payer, whether a health plan, a PBM, or a pharma manufacturer, far more than the per-member fee the product itself charges.

## Cost Drivers

- Model/API cost for the Phase 2 explanation layer.
- Pharmacist review-queue staffing and time.
- Data sync and maintenance cost (DrugBank/openFDA updates).
- Patient and provider support cost.

## Unit Economics View

The B2B2C model only works if the per-member fee stays well below the avoided-cost value it delivers. As a rough sense-check rather than a firm forecast, a PMPM fee in the \$1.50–\$3.00 range is small relative to the cost of even one avoided hospital readmission, which is the logic the whole pricing model leans on. This needs real pilot cost data before it becomes a validated number rather than a planning assumption.

## Scaling Risks

- Latency and cost growth as the GenAI explanation layer scales to more users.
- Workflow complexity as more conditions and languages are added.
- Governance burden as the product expands into new regulatory jurisdictions or the Specialty Pharma Extension.

# 6. Ethics, Governance, and Risk Note

## Main Risks

| Risk                                                 | Likelihood            | Impact      | Mitigation                                                                                                        |
|------------------------------------------------------|-----------------------|-------------|-------------------------------------------------------------------------------------------------------------------|
| Unreviewed high-risk interaction reaches a patient   | Low (with MVP design) | Severe      | Mandatory pharmacist review of all high-risk flags; conservative over-flagging; incident-review process           |
| Regulatory classification as a medical device (SaMD) | Medium                | High        | Built as decision-support with mandatory human review, not autonomous diagnosis; regulatory counsel engaged early |
| Health data privacy breach                           | Low–Medium            | Severe      | Encryption at rest/in transit, minimal retention, HIPAA/DPDP compliance, third-party audit pre-pilot              |
| Model/data drift as new drugs enter the market       | Medium                | Medium      | Scheduled DrugBank/openFDA syncs; unmatched drugs routed to a pharmacist, never silently ignored                  |
| Automation bias: patient over-trust in the app       | Medium                | Medium–High | UX frames the app as a support tool with persistent pharmacist escalation                                         |



## Harm Scenarios

The clearest harm scenario is a genuine high-risk interaction that is missed and reaches a patient without pharmacist review, the single scenario the entire human-in-the-loop design exists to prevent. A second, more social than clinical, harm is over-reliance: a patient treating the app's guidance as a substitute for professional advice in an ambiguous case, or a pharmacist's role being perceived as replaced rather than augmented by the tool.

## Controls

- Guardrails: mandatory pharmacist review of every moderate/high-risk flag; conservative over-flagging rather than under-flagging; unmatched medications are never defaulted to "safe."
- Review process: weekly clinical-quality review of pharmacist-audited alerts; monthly product/growth review; quarterly B2B pilot review.
- Escalation path: any single missed high-risk interaction that reaches a patient without review triggers an immediate incident review, independent of the regular cadence.
- Logging / auditability: every pharmacist decision (approve or dismiss) is logged, giving an auditable trail that health plans and regulators can inspect.

## Launch Blockers

### LAUNCH BLOCKER

If pharmacist agreement with the system's flagged risk level cannot be sustained at or above 95% over the 60-day pilot, or the false-positive rate cannot be kept under 5%, the product should not move beyond that narrow pilot, regardless of engineering readiness elsewhere.